Towards Better Probe-based Steering for Jailbreaking

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Abstract

Recent work has demonstrated the contrastive steering's potential on jailbreaking Large Language Models (LLMs). However, existing contrastive steerings depend on limited contrastive prompts and manually tuned strength, which limits their robustness and effectiveness. In this paper, we leverage the idea of model extraction to refine the steering vectors direction and propose to adaptively tune strength based on training samples' statistics. Experiments demonstrate that our method well improves probe-based steering's effectiveness and robustness, without any extra contrastive prompts or laborious manual tuning. Being an attack paper, this paper focuses on demonstrating the breakdown of fortified LLMs, raising the average harmful score from 5% to 70%.

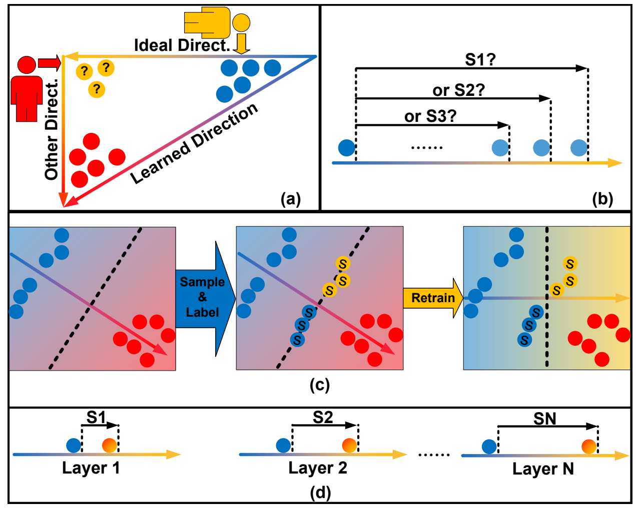

Figure 1. **(a)** The same learned direction, trained on the same contrastive prompts, can form different concept classifiers, which reveals the existence of multiple coupled directions. **(b)** Determining the steering strength requires laborious continuous parameter tuning. To counter these limitations, **(c)** we propose to utilize adaptive retraining to refine the direction, and **(d)** leverage training samples' statistics to determine each layer's steering strength.

1. Introduction

Modern Large Language Models (LLMs) often undergo alignment (Bai et al., 2022; Rafailov et al., 2023) to ensure their harmlessness. However, research (Zou et al., 2023b; Liu et al., 2024; Carlini et al., 2023; Zou et al., 2023a; Xu et al., 2024; Arditi et al., 2024) has shown that such alignment can be circumvented by certain adversarial techniques, known as jailbreaking. To counter jailbreaking, the community has developed various defense methods (Zou et al., 2024; Qi et al., 2025) that reportedly achieve strong adversarial robustness, often claiming near-zero Attack Success Rates (ASRs).

The key challenge in (truly) improving adversarial robustness is developing strong adaptive attacks (Athalye et al., 2018) that can reveal the **worst-case** robustness. When attacking classification models, there exist differentiable proxy (e.g., Cross-entropy Loss) well correlated with the goal such that the gradient-based optimization alone can already form strong adaptive attacks (Madry et al., 2018; Athalye et al., 2018). Yet, when the adversarial goal shifts towards inducing harmful responses from LLMs, such a proxy is not obvious.

So far, at the response level, the community has proposed two types of proxy. The first type is **harmful prefix** (Carlini et al., 2023; Zou et al., 2023b), which aims to increase the probability of responses beginning with harmful tone. The second type is the jailbreaking judge, which directly guides the adversarial attacks (Chao et al., 2025; Mehrotra et al., 2024) to increase the response's harmfulness. While both have been proven effective on certain LLMs, the former may not be well correlated with LLM's harmfulness (Zhou et al., 2025; Liao & Sun, 2024; Zhu et al., 2024), and the latter is usually non-differentiable, hindering its compatibility with the well-developed gradient-based optimization (Zou et al., 2023b; Carlini et al., 2023). These limitations make it difficult to construct powerful automated attacks, even against white-box LLMs.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

One promising technique that can mitigate the above-mentioned limitations is contrastive steering (Zou et al., 2023a; Arditi et al., 2024). Based on the assumption that the concept is represented along a linear direction in the representation space, contrastive steering utilizes the activation of LLMs facing paired contrastive prompts to calculate steering vectors that can control white-box LLM's behavior by at inference time. Besides controlling behavior, the steering vector can also quantify the LLM's behavior, providing a differentiable and behavior-correlated proxy for adversarial attacks (Chen et al., 2025; Huang et al., 2025; Xu et al., 2024). What is more, contrastive steering only requires contrastive prompts **without any corresponding ground truth**, lowering the barrier to use.

In this paper, we push forward the performance of probe-based contrastive steering (Xu et al., 2024; Hedström et al., 2025), a concise and controllable instantiation. All contrastive steering mainly consist of two stages: direction searching and strength tuning. In terms of direction searching, we find that the learned directions contain errors inherently caused by the contrastive prompts. Inspired by model extraction (Tramèr et al., 2016), we propose an algorithm, **without the need for extra contrastive prompts**, that iteratively samples and annotates steered activations to refine the direction searching. Regarding the strength tuning, we find that the commonly used accuracy-based layer selection (Xu et al., 2024) is unstable and that setting layer-uniform logit targets (Xu et al., 2024; Hedström et al., 2025) overlooks the differences in activation magnitudes across layers and thus leads to over-steering. We propose to deprecate the accuracy-based layer selection and use training samples' statistics to guide logit targets setting. Our contributions are as follows:

1. We propose an iterative algorithm to reduce errors in the direction searching stage. The proposed algorithm does not require extra contrastive prompts and only relies on off-the-shelf annotator.

2. We identify the instability of the widely adopted accuracy-based layer selection and the risk of over-steering induced by layer wise-uniform logit targets. We propose to deprecate accuracy-based layer selection and use training statistics to guide logit targets setting, which notably improves the effectiveness and eliminates the need for laborious continuous-parameter tuning.

3. Rather than general-purpose LLMs[1], we mainly conduct experiments on LLMs specifically fortified against jailbreak attempts. The results demonstrate that our method can elevate ASR from near 0% to at least 50%, and mostly to more than 70%.

[1]We leave results on general-purpose LLMs in Appendix D.

2. Preliminary

2.1. Problem Statement

The ultimate goal of studying jailbreaking is to build up reliable LLMs. Formally, given a toxicity-detection oracle isToxic$(\cdot)$, the goal is to solve

$$\min_{\theta} \max_{\Delta\theta} \text{isToxic}(m_\theta(\Delta\theta)) \,, \tag{1}$$

where θ is the LLM's weight, $\Delta\theta$ is any modification that can be applied to the LLM, and $m_\theta(\Delta\theta)$ represents the LLM's state (including but not limited to output texts, hidden states, attention maps, etc.). Here, we refer to the modification as $\Delta\theta$ because both modification in the input space and the embedding space can be seen as altering θ.

The challenge in solving Equation (1) lies in the inner maximization. On one hand, during the training phase, a weak maximization will lead to sub-optimal robustness (Madry et al., 2018). On the other hand, a weak maximization during the evaluation will lead to a false sense of robustness (Athalye et al., 2018). Thus, this paper aims to better solve the inner maximization.

2.2. Autoregressive Decoder-only Transformers

Modern LLMs are mostly autoregressive decoder-only transformers (Vaswani et al., 2017). Vertically, the LLM is composed of stacked decoders with residual connections. Given a hidden state $\mathbf{x}^{(l-1)} \in \mathbb{R}^d$, the l-th decoder $\text{Dec}_l(\cdot)$ conducts

$$\mathbf{x}^{(l)} := \mathbf{x}^{(l-1)} + \text{Dec}_l(\mathbf{x}^{(l-1)}) \,. \tag{2}$$

Horizontally, at each autoregressive step, a L-layer LLM maps the input embedding at the i-th token position $\mathbf{x}_i^{(0)} \in \mathbb{R}^d$ to the last-layer hidden state $\mathbf{x}_i^{(L)}$. $\mathbf{x}_i^{(L)}$ will be later mapped back to an input embedding $\mathbf{x}_{i+1}^{(0)}$ for the next autoregressive step.

2.3. Steering-based Attack

The steering-based attacks are based on the assumption that LLMs' behaviors (or personas) can be controlled by steering the hidden states along a linear direction. Given a L-layer LLM with hidden size d, the steering-based attacks conduct

$$\mathbf{x}^{(l)} := \mathbf{x}^{(l)} + \lambda^{(l)} * \mathbf{v}^{(l)} \,, \tag{3}$$

where $\lambda^{(l)} \in \mathbb{R}$ controls the steering strength at layer l according to $\mathbf{x}^{(l)}$, and $\mathbf{v}^{(l)} \in \mathbb{R}^d$ is the steering vector. As shown in Equation (3), the challenges of designing a steering-based attack lies in **(1)** how to find steering vectors $V = (\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(L)}) \in \mathbb{R}^{L \times d}$ and **(2)** how to determine the steering strength $\lambda^{(l)}$?

2.4. Threat Model

This paper focuses on the worst-case robustness of LLMs against jailbreaking. That is, we only discuss bare LLMs rather than LLM-centered systems (e.g., commercial chatbot) and explore how high the Attack Success Rate (ASR) can be under certain constraint.

Data. In terms of the constraint on data, we assume the attacker only has malicious query set $\mathcal{P}_m$ and benign query set $\mathcal{P}_b$, *without any responses*[2]. Such data restriction, particularly for malicious queries, is sound, as limited attackers usually try to acquire knowledge from the LLM instead of teaching it.

Model. We assume the attacker can access and manipulate all LLM blocks' (e.g., MLP, Self-attention) output during the inference. The attacker can not compute gradients of any LLM's embedding or output value with respect to the LLM's inputs or parameters.

3. Method

In this section, we first formally introduce the assumption on which the steering-based attacks based in Section 3.1. Then, we review how existing methods tackle these two challenges and analyze their limitations in Section 3.2. Lastly, we introduce our method in Section 3.3.

3.1. Linear Control Assumption

The steering-based attacks are based on the assumption that the LLMs' behavior can be controlled by adding vectors to LLMs' hidden state, which is a simple linear operation. Below, to facilitate our discussion of existing methods and ours, we formally describe Linear Control Assumption (LCA).

Definition 3.1 (β-Indicator)**.** β-Indicator $\mathbb{I}_\beta(\cdot) : X \mapsto \{0,1\}$ is an indicator function judging whether the input satisfies a specific behavior (or persona) β, where X is an arbitrary set, and $\mathbb{I}_\beta(x) = \begin{cases} 1, & \text{if } x \text{ satisfies } \beta, \\ 0, & \text{otherwise.} \end{cases}$

Definition 3.2 ((V, H, S)-Similar)**.** Given a vector sequence $V = (\mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(L)})$, a real number sequence $S = (s_1, \ldots, s_L)$, and a similarity metric score$(\cdot,\cdot)$: $\mathbb{R}^d \times \mathbb{R}^d \mapsto \mathbb{R}$, a L-layer LLM's state $m_\theta(\Delta\theta)$ is (V, H, S)-Similar iff. a subset H of its hidden states satisfies $\forall \mathbf{x}_i^{(l)} \in H, \text{score}(\mathbf{x}_i^{(l)}, \mathbf{v}^{(l)}) = s_l$, and is at least (V, H, S)-Similar iff. $\forall \mathbf{x}_i^{(l)} \in H, \text{score}(\mathbf{x}_i^{(l)}, \mathbf{v}^{(l)}) \geq s_l$.

[2] A line of optimized-based steering (Cao et al., 2024; Dunefsky & Cohan, 2025), which should be viewed as a form of parameter-efficient fine-tuning (PEFT), does not fulfill such data constraint because they have ground truth.

Definition 3.2 quantifies the LLM's state. The definition of score$(\mathbf{u}, \mathbf{v})$ depends on the linear model. For example, score$(\mathbf{u}, \mathbf{v}) = \mathbf{u} \cdot \mathbf{v}$ for bare vector sequences, and score$(\mathbf{u}, \mathbf{v}) = \mathbf{u} \cdot \mathbf{v} + b$ if a linear probe with bias b is applied.

Definition 3.3. Given two real number sequences with identical length $A = (a_1, \ldots, a_L)$ and $B = (b_1, \ldots, b_L)$, we say $A > B$, iff. $\forall i, a_i > b_i$, and $A = B$, iff. $\forall i, a_i = b_i$.

Definition 3.3 is introduced to facilitate the comparison of different LLMs' states.

Assumption 3.4. Given a behavior β, there exists an ideal vector sequence $V^* = (\mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(L)})$ such that, for any arbitrary LLM state $m_\theta(\Delta\theta)$, if $m_\theta(\Delta\theta)$ is (V, H, S)-Similar, then $\mathbb{I}_\beta(m_\theta(\Delta\theta))$ is a monotonically increasing function of S on the interval $[\underline{S}, \bar{S}]$.

Remark 3.5 (on Assumption 3.4). If Assumption 3.4 holds, the behavioral control problem of LLMs (i.e., maximizing $\mathbb{I}_\beta(m_\theta(\Delta\theta))$) can be reduced to the problem of increasing the similarity between H and V. The former problem is difficult to solve due to the complex $\mathbb{I}_\beta(m_\theta(\Delta\theta))$, whereas the latter is simple and often admits a closed-form solution, thereby reducing the difficulty of LLM behavioral control.

3.2. Existing Methods

In this section, we review existing steering-based attacks by elucidating the connection between Assumption 3.4 and their design rationale for **direction searching** and **strength tuning**.

3.2.1. Direction Searching

To search V^*, existing methods include two consecutive stages: extracting contrastive hidden states and modeling the difference between contrastive hidden states.

During the extraction stage, given a L-layer LLM with hidden size d, contrastive steering inputs benign instruction set $\mathcal{P}_b$ and malicious instruction set $\mathcal{P}_m$ to capture the LLM's faithful hidden states $H_b^P = \left\{\mathbf{a}_i^{(l)} \mid i \in P, l \in \{1, \ldots, L\}\right\} \subset \mathbb{R}^d$ and faithless hidden states $H_m^P = \left\{\mathbf{b}_i^{(l)} \mid i \in P, l \in \{1, \ldots, L\}\right\} \subset \mathbb{R}^d$ at token positions P, respectively. During the modeling stage, a linear model (including Linear Probes (Xu et al., 2024), PCA (Zou et al., 2023a), Mean-in-Differences (Arditi et al., 2024), etc.) is applied to capture the differences between H_b^P and H_m^P. The outcome of the modeling stage is a vector sequence $V = (\mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(L)})$.

Note that V and the well-aligned LLM normally satisfy $\forall l \in \{1, \ldots, L\}, \mathbb{E}_{i \in P}[\text{score}(\mathbf{v}^{(l)}, \mathbf{a}_i^{(l)})] > \mathbb{E}_{i \in P}[\text{score}(\mathbf{v}^{(l)}, \mathbf{b}_i^{(l)})]$ and $\mathbb{E}_{\Delta\theta \in \mathcal{P}_b}[\mathbb{I}_\beta(m_\theta(\Delta\theta))] > \mathbb{E}_{\Delta\theta \in \mathcal{P}_m}[\mathbb{I}_\beta(m_\theta(\Delta\theta))]$, respectively. From the perspective provided by Assumption 3.4, it thus can be argued that the purpose of the two stages is to find a vector sequence V fulfilling that $\mathbb{I}_\beta(m_\theta(\Delta\theta))$ is a monotonically increasing function of S when $m_\theta(\Delta\theta)$ is (V, H, S)-Similar, **which is the necessary but not sufficient condition of $V = V^*$, due to the non-enumerable $m_\theta(\Delta\theta)$.** Thus, it is hard to claim $V = V^*$.

3.2.2. Strength Tuning

After acquiring the directions, the contrastive steering need to determine the steering strength for each layer. Using the notation from Assumption 3.4, we can say that such strength tuning is about increasing S while determining $\bar{S}$ to bound S. $\bar{S}$ is important because large strength tends to induce incoherent responses. Existing strength tuning can be categorized into constant, ablation, and probes.

Constant. Constant (Zou et al., 2023a) sets $\lambda^{(l)}$ to a fixed value. Constant does not guarantee that $m_\theta(\Delta\theta)$ will be or be at least (V, H, S)-Similar after steering, but trivially increases score$(\mathbf{x}_i^{(l)}, \mathbf{v}^{(l)})$ during each steering. $\bar{S}$ is ignored.

Ablation. Ablation (Arditi et al., 2024; Vu & Nguyen, 2025) sets $\lambda^{(l)} = -\frac{\mathbf{x}_i^{(l)} \cdot \mathbf{v}^{(l)}}{\|\mathbf{v}^{(l)}\|_2^2}$, which will zero out $\mathbf{v}^{(l)}$ in $\mathbf{x}_i^{(l)}$. Ablation guarantees that $m_\theta(\Delta\theta)$ will be $(V, H, \mathbf{0})$-Similar, where $\mathbf{0} \in \mathbb{R}^L$, and assumes $\mathbb{I}_\beta(m_\theta(\Delta\theta)) = 1$ if $m_\theta(\Delta\theta)$ is $(V, H, \mathbf{0})$-Similar.

Probes. When linear probes (Xu et al., 2024; Hedström et al., 2025) $f(\mathbf{x})^{(l)} = \mathbf{w}^{(l)} \cdot \mathbf{x} + b^{(l)}$ are used for modeling, one can guarantee that $m_\theta(\Delta\theta)$ is at least (V, H, S)-Similar by setting $\mathbf{v}^{(l)} = \mathbf{w}^{(l)}$ and $\lambda^{(l)} = \mathbb{I}(s^{(l)} > b^{(l)} + \mathbf{x}_i^{(l)} \cdot \mathbf{w}^{(l)}) * \frac{s^{(l)} - \mathbf{w}^{(l)} \cdot \mathbf{x}_i^{(l)} - b^{(l)}}{\|\mathbf{w}^{(l)}\|_2^2}$. Since linear probes are assumed to well separate H_b^P and H_m^P, it is expected to have $\mathbb{I}(f^{(l)}(\mathbf{x}) \geq 0) \equiv \mathbb{I}_\beta(\mathbf{x})$. $s^{(l)}$ is usually set to a large number to further increase the similarity between V and H.

3.3. Our Method

As previously mentioned, steering-based attacks primarily consist of direction searching and strength tuning, and thus our improvements are mainly focused on these two parts. More specifically, we mainly enhance the probe-based steering, owing to its simplicity and controllability.

3.3.1. Refining Biased Linear Probe with Model Extraction

Supposing that there exists a set of linear probes $f^{*(l)}(\mathbf{x}) = \mathbf{w}^{*(l)} \cdot \mathbf{x} + b^{*(l)}$ that is ideal for steering, we can view the direction searching as model extraction (Tramèr et al., 2016) because the goal of direction searching is to obtain a set of

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Algorithm 1 Iterative Training Set Augmentation with Contrastive Activations

Require: Contrastive prompts $\mathcal{P}_m$ and $\mathcal{P}_b$, LLM $m(\cdot)$ with L layers, Annotator $\mathbb{I}_{faithful}(\cdot)$ returning $(data, label)$, Total iterations T, Token Position P, Steering strength $S = \{s^{(l)} \mid l \in \{1, 2, \ldots, L\}\}$

Ensure: Final trained probe sets F_T

```
A_0 ← I_faithful(m({P_m, P_b, P}))
Train initial probe sets F_0 using A_0
for i = 0 to T − 1 do
    A_aug ← I_faithful(m({P_m, F_i, P}))
    A_{i+1} ← A_i ∪ A_aug
    Train F_{i+1} using A_{i+1}
end for
return F_T
```

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linear probes $f^{(l)}(\mathbf{x})$ satisfying $f^{(l)}(\mathbf{x}) \approx f^{*(l)}(\mathbf{x})$, similar to model extraction. Although $f^{*(l)}(\mathbf{x})$ is inaccessible, since it is expected that $\mathbb{I}(f^{*(l)}(\mathbf{x}) \geq 0) \equiv \mathbb{I}_\beta(\mathbf{x})$, we can use $\mathbb{I}_{faithful}(\mathbf{x})$ as a proxy, which can be any reliable jailbreaking judge (Souly et al., 2024) in practice.

From this perspective, contrastive steering is inherently retraining-based model extraction (Tramèr et al., 2016): annotate activations with the judge, and train a linear model with annotated activations. Such perspective inspires us to refine the linear model by including more annotated activations. One simple approach is using more contrastive prompts. However, **specifically for the jailbreaking task we discuss**, since we can hardly get harmful prompts that can be followed by the aligned LLM, the contrastive prompts must be formed by harmful and harmless prompts, which contain other coupled concept directions (e.g., ethical-to-unethical, or even cake-to-bomb) other than the desired faithful-to-faithless direction. Thus, the extracted model is inherently $\mathbb{I}_{faithful}(\mathbf{x}) * \mathbb{I}_{Noise}(\mathbf{x})$. Here, we argue that $\mathbb{I}_{Noise}(\mathbf{x})$ is practically unignorable because one can train the desired faithfulness judge as well as an ethic judge **with the same harmful-harmless contrastive prompts.**

To reduce the noise induced by limited data and contrastive prompts, we propose to iteratively augment the training set with more contrastive activations that are annotated by reliable $\mathbb{I}_{faithful}(\mathbf{x})$ and correspond to **same prompts**. Specifically, after we obtain the i-th probe sets $F_i = \{f_i^{(l)}(\mathbf{x}) \mid l \in \{1, 2, \ldots, L\}\}$, we input harmful prompts, generate responses, and collect activations with the LLM steered by F_i. Then, we annotate each activation by judging their corresponding responses with $\mathbb{I}_{faithful}(\mathbf{x})$, and add these annotated activations to the training set for next iteration. The pseudo code is presented in Algorithm 1.

The proposed iterative algorithm has only one hyper-

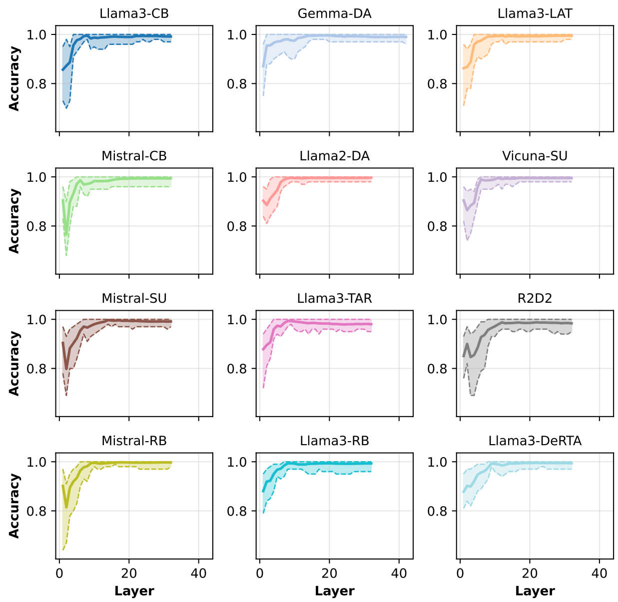

Figure 2. Linear probes' accuracy on validation activations across layers. The line represents the mean accuracy, and the shaded area indicates the max and min values of accuracy over 50 random samplings.

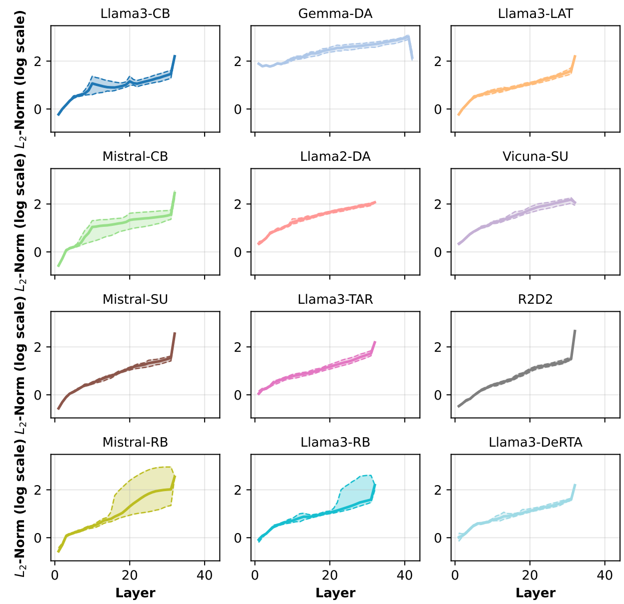

Figure 3. L_2-norm of activation across layers. The line represents the mean norm, and the shaded area indicates the max and min values of the norm for 100 different activation.

parameter: The steering strength $S = (s_1, \ldots, s_L)$. We set $s^{(l)} = 0$, which steers activations to just cross the decision boundary. The reason is that augmenting the training set with the points that current classifier is least certain about has been proven to be more effective and efficient than with random sampled points, which is known as **adaptive retraining** (Tramèr et al., 2016) or active learning (Cohn et al., 1994).

3.3.2. Improving and Simplifying Strength Tuning

Strength tuning can be laborious hyper-parameter tuning process since, given a L-layer LLM, there are L continuous parameters to be determined. Existing probe-based steering (Xu et al., 2024; Hedström et al., 2025) sets layer-uniform $s^{(l)}$ significantly larger than 0 to ensure altered behavior. Xu et al. (2024) assumes that $f^{(l)}(\mathbf{x})$ having low test accuracy suggest that LCA does not hold at layer l. Consequently, they further only steer layers (usually those close to the output layer) having test accuracy above certain threshold (e.g., 90%)[3].

However, we point out that such accuracy-based selection is not robust. We randomly sample 100 paired examples from the contrastive prompts provided by Arditi et al. (2024), split them equally into training and test sets (50% each), and train linear probes. This procedure is repeated 50 times. As shown in Figure 2, the test accuracy is unstable, especially for early layers. Since whether LCA is hold at certain layer is deterministic and objective, we believe that such accuracy-based perspective is unreliable. Instead, we argue that the exact layerwise discrepancy should be considered is activations' magnitude. We sample 100 prompts and calculating the L_2-norm of activations per layer. In Figure 3, we can find that the activation's magnitude increases by orders of magnitude from early layers to late layers. For probe-based steering, when we steer $\mathbf{x}_i^{(l)}$ and set steering strength to $s^{(l)}$, the ratio of the steering vector's norm to $\|\mathbf{x}_i^{(l)}\|_2$ is

$$\frac{s^{(l)} - \mathbf{w}^{(l)} \cdot \mathbf{x}_i^{(l)} - b^{(l)}}{\|\mathbf{w}^{(l)}\|_2 * \|\mathbf{x}_i^{(l)}\|_2} . \tag{4}$$

Clearly, with a layer-uniform $s^{(l)}$, over-steering occurs when $\|\mathbf{x}_i^{(l)}\|_2$ is too small.

Given the limitations described above, we propose to set $s^{(l)}$ according to $\|\mathbf{x}_i^{(l)}\|_2$. One simple and adaptive approach is utilizing the logits of activation $\mathbf{y}_i^{(l)}$ corresponding to the desired behavior. Intuitively, if the probe is ideal, setting the target as the logits of $\mathbf{y}_i^{(l)}$ will guarantee the steered LLM exhibiting the same behavior as the LLM corresponding to $\mathbf{y}_i^{(l)}$. In terms of the activation's magnitude

[3] Such an idea is also implicitly adopted in other types of steering. For example, (Zou et al., 2023a) manually choose middle layers for steering.

we concern, when we set $s^{(l)} = \mathbf{w}^{(l)} \cdot \mathbf{y}_i^{(l)} + b^{(l)}$, the ratio of the steering vector's norm to $\|\mathbf{x}_i^{(l)}\|_2$ becomes

$$\frac{\|\mathbf{y}_i^{(l)}\|_2 * \cos\theta_{wy} - \|\mathbf{x}_i^{(l)}\|_2 * \cos\theta_{wx}}{\|\mathbf{x}_i^{(l)}\|_2}. \tag{5}$$

Furthermore, since the norm of same-layer activations are of similar order, assuming $\|\mathbf{y}_i^{(l)}\|_2 \approx \|\mathbf{x}_i^{(l)}\|_2$, we can reduce Equation (5) to $\cos\theta_{wy} - \cos\theta_{wx}$, a value independent of activations' magnitude but depended on the direction.

3.3.3. Other Implementation Details

Discarding the Last-layer Activation. Though also named hidden states in most implementation, the last decoder's activation is quite different from others. First, steering the last decoder's activation is equivalent to setting logit bias, which can easily induce repetition. Second, the influence of the last decoder's activation is local rather than contextual: It isn't involved in latter autoregressive steps' calculation but only influences the current steps' unembedding. Given these reasons, we choose to ignore the last decoder's activation, which is a common setting proactively adopted by Zou et al. (2023a), Vu & Nguyen (2025) and Arditi et al. (2024), and by Xu et al. (2024) through bug[4].

Steering All Token Positions. Since the activations used for direction searching is usually from response token position, some prior works (Xu et al., 2024; Chen et al., 2025) steer only response tokens. Yet, if we view the steering vectors as a part of LLM's architecture and weight, it is trivial that we should steer activations of all token positions. In particular, probe-based steering can be seen as a rank-1 LoRA (Hu et al., 2022) with fixed bias. Given an activation $\mathbf{x}_i^{(l)} \in \mathbb{R}^{d\times 1}$, probe-based steering can be written as

$$\mathbf{x}_i^{(l)} := \underbrace{\mathbf{I}_d \mathbf{x}_i^{(l)}}_{W_0 x} + \underbrace{(-\hat{\mathbf{w}}^{(l)} \hat{\mathbf{w}}^{(l)\intercal} \mathbf{x}_i^{(l)})}_{BAx} + \underbrace{\frac{s^{(l)} - b^{(l)}}{\|\mathbf{w}^{(l)}\|_2^2} * \mathbf{w}^{(l)}}_{\text{Fixed Bias}}, \tag{6}$$

where $\hat{\mathbf{w}}^{(l)} = \frac{\mathbf{w}^{(l)}}{\|\mathbf{w}^{(l)}\|_2}$. We argue that we should treat the steering vector as a normal parameter efficient adapter, and thus applying all-position steering is crucial.

4. Experiment

4.1. Setups

Datasets. We use the first 100 pairs of contrastive prompts provided by Arditi et al. (2024) for direction searching. We use the first 100 harmful prompts of StrongReject (Souly et al., 2024) and of Harmbench (Mazeika et al., 2024) labeled "Standard", resulting in 200 harmful prompts for evaluation.

Models. We mainly focus on LLMs specifically fortified against jailbreaking. We include Circuit Breaker (CB) (Zou et al., 2024), Deep Alignment (DA) (Qi et al., 2025), RepBend (RB) (Yousefpour et al., 2025), R2D2 (Mazeika et al., 2024), SafeUnlearning (SU) (Zhang et al., 2024), TAR (Tamirisa et al., 2025), DeRTA (Yuan et al., 2025) and Latent Adversarial Training (LAT) (Sheshadri et al., 2025), resulting in 12 fortified LLMs in total.

Metrics. All judges we use are LLM-based, including Fine-tuned LLM (SRF) provided by Souly et al. (2024), a rubric judge (SR) powered by StrongReject's prompt (Souly et al., 2024) and Qwen-Plus (Yang et al., 2025), and the HarmBench classifier (HB) (Mazeika et al., 2024). **We employ these three judges because their design and training emphasize the helpfulness of responses, rather than focusing solely on refusal.** All judges output scores ranging from 0 to 1. The maximum length of responses during evaluation is 512. **Since we use SRF for direction searching, we believe using more judges than SRF is crucial for mitigate potential biases (an effect similar to reward hacking) in evaluation.**

Baselines. The baselines we compare are all steering-based attacks. We include RepE (Zou et al., 2023a), Refusal Direction (RD) (Arditi et al., 2024) that has both constant-based (RD-C) and ablation-based (RD-A) strength tuning, SCAV (Xu et al., 2024), and Angular Steering (Vu & Nguyen, 2025). **All baselines use the same data as ours.** SCAV steers only prompt token positions, while others steer all token positions.

Hyper-Parameters. During the direction searching, we set $s^{(l)} = 0$ and $T = 20$. The token position we choose is the position immediately preceding the first response token, a common setting adopted by prior works (Zou et al., 2023a; Xu et al., 2024; Vu & Nguyen, 2025). We split the 100 pairs contrastive prompts into training set and validation set in 50:50 ratio. 50 pairs are used for Algorithm 1, and the other 50 harmful prompts are used for validation to pick the best direction. The rest 50 benign prompts are of no use. The maximum responses length during the direction searching (including validation) is 256. After weighing cost, accuracy and flexibility, we choose SRF as the annotator, wherein samples scored less than 0.05 is classified as negative and scored larger than 0.6 as positive. The rest is discarded, given the relative noisy scoring. Class balance is applied for probe training. During the inference (including validation), for each layer, we choose the largest known

[4] Xu et al. (2024) mistakenly takes activations from layer l as layer $l+1$ and consequently discard the last decoder's. The code we adopt in this paper is corrected.

Table 1. The performance of contrastive steering against 12 different fortified LLMs. * means no available direction is found. The best result of each column us in **bold**.

Method	Llama2-DA			Gemma-DA			Vicuna-SU			Mistral-SU			Mistral-RB			Llama3-RB			Llama3-LAT			Llama3-TAR			Mistral-CB			Llama3-CB			R2D2			Llama3-DeRTA			Avg.
	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	
RepE	0.01	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.02	0.02	0.00	0.05	0.04	0.00	0.04	0.03	0.00	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.02	0.01	0.01	0.02	0.04	0.04	0.01	0.01	0.05	0.00	0.02
SCAV	0.01	0.02	0.01	0.37	0.43	0.57	0.01	0.03	0.01	0.01	0.01	0.00	0.01	0.04	0.00	0.00	0.00	0.01	0.01	0.03	0.01	0.01	0.01	0.01	0.02	0.02	0.00	0.01	0.03	0.01	0.01	0.05	0.01	0.01	0.02	0.00	0.05
RD-A	*	*	*	0.58	0.56	0.78	*	*	*	0.08	0.04	0.15	0.08	0.04	0.11	0.64	0.71	0.91	0.14	0.17	0.21	0.18	0.14	0.29	*	*	*	0.09	0.12	0.11	0.23	0.29	0.44	0.31	0.42	0.49	0.31
RD-C	*	*	*	0.36	0.50	0.54	*	*	*	0.04	0.02	0.08	0.21	0.20	0.32	0.28	0.43	0.44	0.08	0.11	0.14	0.14	0.14	0.23	*	*	*	0.37	0.45	0.54	0.18	0.26	0.38	0.07	0.12	0.16	0.25
Angular	0.03	0.07	0.03	0.23	0.19	0.27	0.01	0.00	0.00	0.03	0.00	0.08	0.08	0.02	0.13	0.06	0.01	0.08	0.07	0.07	0.10	0.26	0.20	0.41	0.02	0.02	0.02	0.02	0.05	0.02	0.22	0.30	0.40	0.28	0.38	0.43	0.13
Ours	**0.63**	**0.91**	**0.89**	**0.69**	**0.71**	**0.90**	**0.61**	**0.74**	**0.84**	**0.43**	**0.55**	**0.71**	**0.68**	**0.73**	**0.89**	**0.71**	**0.78**	**0.96**	**0.69**	**0.80**	**0.95**	**0.35**	**0.25**	**0.53**	**0.71**	**0.73**	**0.93**	**0.71**	**0.82**	**0.91**	**0.26**	**0.36**	**0.60**	**0.58**	**0.72**	**0.88**	**0.70**

Table 2. The performance gain of components proposed for strength tuning.

Method	Llama2-DA			Gemma-DA			Vicuna-SU			Mistral-SU			Mistral-RB			Llama3-RB			Llama3-LAT			Llama3-TAR			Mistral-CB			Llama3-CB			R2D2			Llama3-DeRTA			Avg.
	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	
SCAV	0.01	0.02	0.01	0.37	0.43	0.57	0.01	0.03	0.01	0.01	0.01	0.00	0.01	0.04	0.00	0.00	0.00	0.01	0.01	0.03	0.01	0.01	0.01	0.01	0.02	0.02	0.00	0.01	0.03	0.01	0.01	0.05	0.01	0.01	0.02	0.00	0.05
SCAV+DLA+SAT	0.01	0.04	0.01	0.36	0.46	0.57	0.01	0.02	0.01	0.01	0.02	0.00	0.01	0.03	0.00	0.01	0.02	0.01	0.01	0.03	0.01	0.01	0.01	0.01	0.02	0.02	0.00	0.01	0.03	0.01	0.01	0.05	0.01	0.01	0.02	0.00	0.05
SCAV+AS	0.07	0.33	0.13	0.47	0.48	0.68	0.44	0.54	0.78	0.01	0.04	0.01	0.01	0.01	0.00	0.01	0.02	0.01	0.01	0.02	0.01	0.11	0.07	0.23	0.01	0.01	0.01	0.01	0.03	0.01	0.01	0.01	0.01	0.05	0.16	0.12	0.14
SCAV+AS+DLA	0.11	0.45	0.16	0.47	0.48	0.66	0.44	0.54	0.74	0.24	0.21	0.46	0.18	0.29	0.35	0.17	0.16	0.28	0.55	0.72	0.83	0.10	0.04	0.20	0.07	0.10	0.16	0.03	0.10	0.06	0.10	0.10	0.23	0.16	0.31	0.33	0.29
SCAV+AS+DLA+SAT	0.07	0.39	0.08	0.46	0.35	0.68	0.59	0.77	0.88	0.40	0.45	0.64	0.30	0.40	0.61	0.59	0.72	0.90	0.54	0.63	0.85	0.10	0.04	0.23	0.24	0.20	0.54	0.15	0.22	0.29	0.19	0.17	0.40	0.08	0.21	0.15	0.40

logit of training samples as the steering strength. Formally, we set $s^{(l)} = \max_{x^{(l)} \in H^{(l)}_{train}} f^{(l)}(\mathbf{x}^{(l)})$.

4.2. Colosseum

Comparison between Attacks. As presented in Table 1, our methods exhibits better effectiveness and robustness than others. When attacking Gemma-DA, Llama3-RB, and Llama3-CB, which have already shown notable vulnerability to other contrastive steering, our method can further notably raise the harmful scores on all metrics. In terms of robustness, our method achieves SR scores of at least 0.53 on 12 LLMs, with most reaching at least 0.8, while others exhibit both SR scores larger than 0.4 and close to 0 across different LLMs. Specifically, RepE achieves all near zero performance. SCAV has near zero scores on most LLMs except Gemma-DA. According to Figure 3, we believe the reason is that Gemma-DA has activation of large magnitude than others, which can tolerate the large scale steering recommended by Xu et al. (2024). RD exhibits the second-best effectiveness on average, yet also fails to find available direction on several LLMs due to its direction filtering.

Comparison between LLMs. Among these fortified LLMs, Llama3-TAR (Tamirisa et al., 2025) exhibits the best robustness. We believe this is because TAR conducts adversarial training over fully tampered parameters while contrastive steering can be seen as tampering only partial parameters. As a comparison, Llama3-LAT conducts adversarial training over the same activations as contrastive steering, and thus more vulnerable than Llama3-TAR. The SU-series counters jailbreaking by unlearning harmful knowledge, Yet, without any ground truth, our method successfully elicits harmful responses, indicating that the SU-series still contain harmful knowledge. RB- and CB-series defend against jailbreaking by manipulating activations, wherein CB-series was reported that perturbing activations (i.e., applying RepE) can not bypass it (Zou et al., 2024). Our method, as well as RD, show that perturbing activations can jailbreak CB-series [5]. Bypassing DA-series and DeRTA, which both tackle shallow alignment, indicates that the contrastive steering does not depend on shallow alignment (Qi et al., 2025). R2D2 exhibits the second-best robustness, yet we will show that such robustness is induced by degraded capability.

4.3. Ablation Study

In this section, we analyze the four components of our method. Unlike the typical ablation studies, we do not focus on evaluating the effectiveness of these four components, but rather on their indispensability. Our four components can be categorized into strength tuning and direction searching.

4.3.1. Strength Tuning

Let us use the probes trained on initial contrastive activations to investigate how our strength tuning strategy benefits the probe-based steering.

Adaptive Strength (AS). The adaptive strength is essential. Without it, even if the searched direction is ideal, we can hardly tackle over-steering unless we conduct laborious grid search over L continuous parameters. When setting $s^{(l)} = \sigma^{-1}(99.99\%)$ and use the validation accuracy to determine steer or not, as proposed by Xu et al. (2024), we can only get harmful scores near zero, as shown in Table 2 and row 2, Table 1. The only exception is Gemma-DA. The reason is that Gemma-DA's activation magnitude is large such that setting $s^{(l)} = \sigma^{-1}(99.99\%)$ will not cause over-steering. After applying AS, as presented in Table 2, we successfully drag Llama2-DA, Vicuna-SU, Llama3-TAR and Llama3-DeRTA out of the zero zone, while DLA+SAT alone can not.

[5] Xu et al. (2024) reported that SCAV can bypass CB-series. Yet, it can not be reproduced (even by the authors).

Table 3. The performance gain of components proposed for direction searching. * means no available direction is found.

Method	Llama2-DA			Gemma-DA			Vicuna-SU			Mistral-SU			Mistral-RB			Llama3-RB			Llama3-LAT			Llama3-TAR			Mistral-CB			Llama3-CB			R2D2			Llama3-DeRTA			Avg.
	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	
Rep+NA	0.01	0.00	0.00	0.01	0.00	0.01	0.01	0.00	0.02	0.01	0.00	0.05	0.04	0.00	0.08	0.02	0.01	0.09	0.00	0.00	0.00	0.09	0.09	0.14	0.01	0.02	0.02	0.01	0.02	0.01	0.04	0.06	0.06	0.01	0.05	0.01	0.03
SCAV+NA	0.04	0.13	0.07	0.29	0.29	0.29	0.03	0.25	0.03	0.01	0.04	0.01	0.01	0.02	0.01	0.01	0.03	0.01	0.01	0.03	0.01	0.06	0.06	0.11	0.01	0.04	0.01	0.01	0.03	0.01	0.01	0.01	0.01	0.01	0.02	0.01	0.06
RD-A+NA	*	*	*	0.62	0.60	0.83	*	*	*	0.13	0.08	0.25	0.26	0.17	0.35	0.67	0.72	0.91	0.11	0.14	0.16	*	*	*	*	*	*	0.15	0.18	0.21	0.25	0.29	0.41	0.09	0.14	0.19	0.33
RD-C+NA	*	*	*	0.38	0.48	0.57	*	*	*	0.05	0.01	0.12	0.28	0.32	0.46	0.25	0.38	0.41	0.10	0.13	0.14	*	*	*	*	*	*	0.17	0.26	0.25	0.28	0.40	0.49	0.06	0.12	0.13	0.26
Angular+NA	0.11	0.09	0.14	0.32	0.28	0.44	0.10	0.09	0.19	0.14	0.11	0.27	0.16	0.12	0.28	0.03	0.04	0.04	0.14	0.14	0.20	0.27	0.19	0.38	0.03	0.03	0.02	0.01	0.04	0.03	0.21	0.25	0.43	0.36	0.55	0.58	0.19
SCAV+AS+DLA+SAT+NA	0.19	0.49	0.34	0.52	0.51	0.77	0.52	0.65	0.75	0.25	0.23	0.40	0.32	0.35	0.59	0.58	0.69	0.87	0.56	0.67	0.88	0.11	0.06	0.22	0.16	0.11	0.30	0.39	0.45	0.70	0.16	0.23	0.34	0.33	0.48	0.66	0.44
SCAV+AS+DLA+SAT+AR	0.63	0.91	0.89	0.69	0.71	0.90	0.61	0.74	0.84	0.43	0.55	0.71	0.68	0.73	0.89	0.71	0.78	0.96	0.69	0.80	0.95	0.35	0.25	0.53	0.71	0.73	0.93	0.71	0.82	0.91	0.26	0.36	0.60	0.58	0.72	0.88	0.70

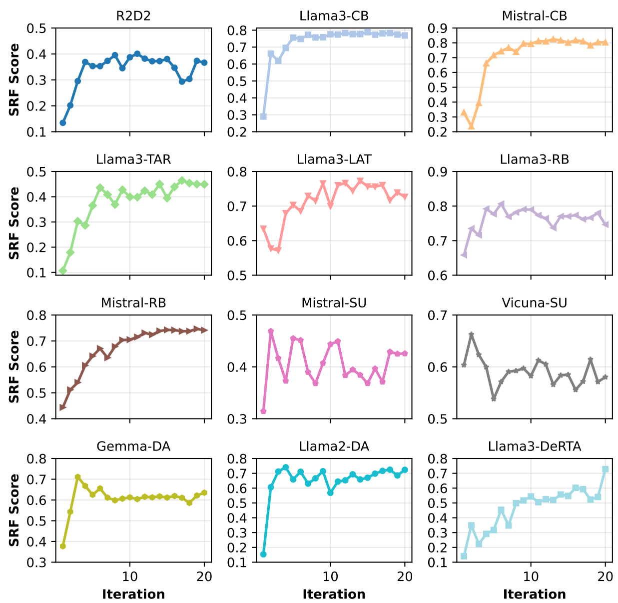

Figure 4. The SRF Score of steered LLMs across iterations on the validation set.

Discarding the Last-layer Activation (DLA). Steering the last-layer activation is equivalent to applying logit bias. While such steering may somehow utilize the shallow alignment (Qi et al., 2025), jailbreaking the LLM by forcing it to start with harmful tone, such steering has no benefits to coherence. In Table 2, we can find that discarding the last-layer activation further improves the harmful scores.

Steering All Tokens (SAT). As mentioned in Section 3.3.3, the probe-based steering can be viewed as a form of parameter efficient adapter and should be applied to all token positions if so. As shown in Table 2, SAT improves performance by 11% on average. Notably, on CB- and RB-series, it raises harmful score by a large margin. We believe the reason is that CB and RB also manipulate activations of prompt token positions during their training. If only response tokens are steered, the over all generation is still based on manipulated prompt tokens, leading to unsuccessful steering[6].

4.3.2. Model Extraction.

So far, we have shown how AS, DLA, and SAT improve and simplify probe-based steering. Yet, with biased directions, the adaptive logit target based on samples' statistic may correspond to biased behavior. Below, we investigate how our model extraction view benefit direction searching

Naive Augmentation (NA). Adding more contrastive prompts is the simplest approach to sample more annotated activations. We use all 871 harmful prompts and the first 871 benign prompts from Arditi et al. (2024) to extract activations and train probes, resulting in 1,742 training samples, which is more than our 20-iteration adaptive retraining has (1,100 at most). In Table 3, we can find that NA only marginally improves performance by at most 6% on average. This result is accord with our insight that the noised direction, coupled with the desired direction, will hinder the effectiveness of NA.

Our Adaptive Retraining (AR). As shown in Table 3, our AR improves probe-based steering by 30% on average and on 11 LLMs out of 12. These results demonstrate AR's competitive effectiveness and data efficiency. We also present the SRF scores on the validation set across iterations in Figure 4. We can find that the porbe-based steering's performance is improved over iterations and fluctuates within a certain range when the AR is converged. Such tendency is obvious on LLMs other than the SU-series. We hypothesize that, under the universal hyper-parameters, AR may have already converged on SU-series, leading to such fluctuation.

5. Conclusion

By utilizing training samples' statistics and adaptive retraining, we successfully improved the probe-based steering's robustness and effectiveness. We demonstrated that our method can improve the harmfulness of LLMs that have already shown vulnerability to prior steering and can reveal the vulnerability of LLMs that prior steering did not reveal. We hope that our method can benefit the robustness evaluation (red-teaming), which is the foundation of developing truly reliable defenses. We leave discussions on limitations and future works in Appendix A.

[6]More probably than not, some early versions of SCAV might steer all tokens and thus bypass the CB-series. Yet, after observing degradation on other LLMs, SCAV might deprecate SAT.

Impact Statement

While the proposed algorithm may be directly utilized to jailbreak open-source LLMs, we believe it can enhance the effectiveness and robustness of red-teaming, which is essential for building up reliable defenses. Also, since the proposed algorithm is evaluated on eliciting faithful persona, we believe, if such persona control is not special, such algorithm can also be applied to control other personas of the LLM for good use.

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Table 4. The performance of our method with different SRF thresholds.

Method	Llama2-DA			Gemma-DA			Vicuna-SU			Mistral-SU			Mistral-RB			Llama3-RB			Llama3-LAT			Llama3-TAR			Mistral-CB			Llama3-CB			R2D2			Llama3-DeRTA			Avg.
	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	SRF	HB	SR	
Ours (0.05, 0.6)	0.63	0.91	0.89	0.69	0.71	0.90	0.61	0.74	0.84	0.43	0.55	0.71	0.68	0.73	0.89	0.71	0.78	0.96	0.69	0.80	0.95	0.35	0.25	0.53	0.71	0.73	0.93	0.71	0.82	0.91	0.26	0.36	0.60	0.58	0.72	0.88	0.70
Ours (0.05, 0.8)	0.66	0.86	0.92	0.77	0.78	0.93	0.59	0.74	0.87	0.42	0.56	0.69	0.60	0.67	0.89	0.71	0.90	0.96	0.70	0.81	0.96	0.32	0.23	0.51	0.67	0.66	0.87	0.66	0.69	0.93	0.32	0.41	0.66	0.55	0.74	0.87	0.70
Ours (0.025, 0.7)	0.62	0.84	0.85	0.66	0.60	0.84	0.62	0.75	0.86	0.43	0.50	0.69	0.63	0.69	0.86	0.70	0.83	0.96	0.71	0.85	0.96	0.34	0.23	0.46	0.70	0.62	0.87	0.68	0.75	0.96	0.27	0.38	0.64	0.48	0.65	0.80	0.68

A. Limitations and Future Work

Efficiency in practice. The main limitation of our methods is its time cost, which is mainly caused by generating response for annotation. During the direction searching, we set the maximum response length to 256 such that SRF can judge responses accurately enough while the time induced by generation is acceptable. We believe such limitation can be trivially mitigated by more advanced GPU, more efficient inference framework, and more powerful annotators that can judge shorter truncated responses.

Application to Controlling Other Personas or Behaviors. This paper focuses on jailbreaking, wherein the faithfulness is the persona we concern. If such persona is not special, we believe our method can be utilized to control other personas or behaviors. Yet, as we identified, LCA determines the theoretical feasibility, and, to date, the annotator's reliability determines the practical feasibility.

Application to Reasoning Models. Reasoning models are known for their lengthy CoT. Given that current annotators even struggle to accurately judge instructed model's responses, we can hardly apply our method to reasoning models.

B. Exhaustive Annotator Threshold Tuning

Currently, the most exhausting part of our method is the annotator.

The annotator is the proxy of ideal probes but is inevitably noised in practice. Regarding the SRF we use, a faithless but detailed disclaimer may be scored 0.4 while the beginning of a truncated faithful lengthy response may be scored only 0.2. Therefore, we have to set a low threshold and a high threshold that are far apart, to avoid the ambiguous score interval. However, such threshold setting may discard considerable generated responses, leading to inefficiency. Consequently, our universal setting, 0.05 and 0.6, is a result weighing accuracy and efficiency. Note that such universal setting may not be the optimal setting for each LLMs. As shown in Table 4, the best SRF thresholds for each LLM varies, and the thresholds, 0.05 and 0.6, we use by default is definitely not the optimal.

Using close-source LLMs as rubric annotator might reduce the need for exhausting threshold tuning, but close-source LLMs often exhibit randomness (He & Lab, 2025), making

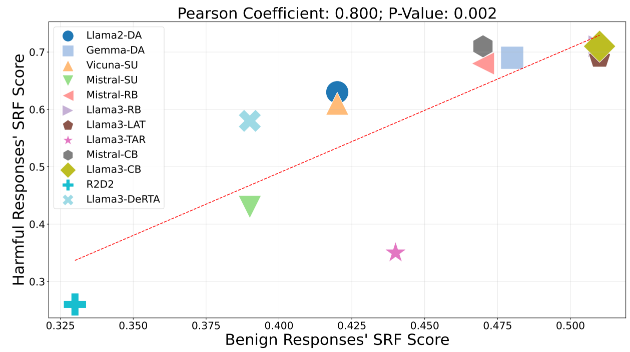

Figure 5. Correlation between benign responses' SRF score and harmful responses' SRF score across different LLMs.

them even harder to control and making SRF with thresholds the best annotator we know of in terms of efficiency, accuracy, and controllability. Currently, we do not bother and not recommend conducting grid search on these thresholds because tuning thresholds will not improve the underlying annotator but reduce misclassified samples by sacrificing efficiency. Instead, it is more worthwhile to build a stronger new annotator.

Note that the above issues are not inherent limitation of our method. We believe that powerful annotators in the future (or expensive human annotators at present) will trivially free our method from such exhaustive threshold tuning.

C. Capability and Safety

In Table 1, R2D2 exhibits the second-best harmlessness. Yet, we find that such harmlessness is induced by degraded overall helpfulness. Using SRF to judge helpfulness, we evaluate all 12 LLMs' responses quality. In Figure 5, we can find a clear positive correlation between the helpfulness of harmful responses and benign responses. Notably, R2D2 has the worst helpfulness (We manually check R2D2's responses. Even if the prompt is benign, it still tends to response with "I am not able to do..."). That is to say, if the LLM can not properly follow benign prompts, it can thus hardly follow harmful ones, even if it is willing to follow.

While, in practice, one can ignore poor LLMs and choose to jailbreak advanced LLMs for high-quality responses, how to deal with LLMs that consistently have poor attitudes remains interesting. Let us take R2D2 as an example. First, observing the clear positive correlation between the helpfulness of harmful responses and benign responses, we filter out benign prompts that R2D2 can not (or refuse to) follow, i.e., those with SRF score lower than 0.5.

D. Results on General-purpose LLMs